LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	24
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Rate sensing about three axes, single axis rate sensing, combined use of rate and displacement gyros.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Rate sensing about three axes, single axis rate sensing, combined use of rate and displacement gyros.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagrams and gyro axis configurations; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three axes of rotation for an aircraft?
(b) Define the function of a displacement gyro.
(c) How does a rate gyro differ from a displacement gyro?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Rate sensing about three axes • Single axis rate sensing principles • Gyroscopic rigidity in motion	L & Ds	PPT, IB	15 min
(b)	• Combined use of rate and displacement gyros • AFCS integration logic • Error compensation mechanisms	L & Ds	PPT, PDF	20 min
(c)	• Practical application of AFCS • System limitations and constraints • Maintenance implications	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how rate gyros contribute to automatic flight control stability.
(b) Describe the interaction between rate and displacement gyros in the AFCS.
(c) Identify one limitation of single-axis rate sensing in multi-axis maneuvers.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of rate sensing discussed in this lesson apply to the leveling and compensation systems covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of leveling and compensation system.	Refer to the master syllabus reference list	LP 25

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min